

everything for them; they can just focus on the application they are writing and leverage the data the application is storing. The autonomous technology takes care of all database management headaches, right from setting it up to patching it and upgrading it.

Beyond developers and IT teams, business users are realising enormous benefits as well by better extracting value from data and improving decision making. Let me give you an interesting customer example of ecommerce product development startup Codeinks. This Bengaluru based company used Oracle Autonomous Data Warehouse out-of-the-box to analyse heavy-duty ecommerce analytics workloads, and immediately realised an average performance improvement of 30-50 per cent for their defined scenarios.

Another trend that I see is developers being very open to experimenting with emerging technologies in their applications. We are no longer talking about simple Web applications, but about transformational apps which embed IoT, the blockchain for transactions across multiple organisations that do not trust each other fully, and look at how the power of AI/ML can be better applied to the business.

Q Can you elaborate on the open source solutions or projects that you are working on.

A: We are doing a lot of work in the open source space. Oracle Linux is a prime example. An additional bit of information that not many are aware of is that we provide a specialised version of Oracle Linux Kernel for customers running Oracle workloads. If you go a little higher on the value chain, then we have Oracle MySQL, which is the world's most popular open source database. We also are working on an open source serverless engine, called Project Fn. This framework enables developers to code and test locally, and then deploy their work on any cloud. We also contribute to Hyperledger Fabric, which is the basis of our blockchain offering.

Q What are the emerging technologies that developers are working on and how do you expect these to change the industries of tomorrow?

A: These are exciting times to be a developer. Today, we are no longer talking about simple applications. The industry is already changing, and I see a lot of differences just in terms of the applications and the solutions that we are building. We are talking about complex use cases, which bring together multiple transformational technologies, such as IoT, blockchain, etc.

We see a lot of organisations in financial services, manufacturing, pharma, healthcare, etc, looking at how to improve the efficiency of their transactions that are happening between parties that have only partial mutual trust. Of late, we're seeing blockchain projects quickly moving from pilot to production, as enterprises and various government entities begin to see the inherent value of distributed ledgers and smart contracts. With the Oracle Blockchain Cloud service, companies can effectively manage their transactions and trace the ownership.

Another trend that is very visible is that AI/ML is becoming embedded as an integral part of cloud applications. For instance, our Software as a Service (SaaS) offerings, such as the enterprise resource planning (ERP) and human capital management (HCM) solutions, are embedded with AI/ML, by default. Further, the proliferation of virtual assistants such as Alexa, Siri etc, is making conversational AI a raging trend. So I anticipate that soon, devices will come with AI/ML embedded, by default, as developers figure out newer ways of interacting with enterprises' applications. We have created the Oracle Digital Assistant, with natural language processing (NLP) and AI/ML built in. This assistant goes well beyond the standard chatbots available today that provide simple, single-skilled, linear responses.

Q What is the future road map for the Oracle Cloud Platform?

A: Our focus for the future will be mainly to build further on our

autonomous offerings, to see how we can enhance them further to make them more useful to customers and even more widely available. Second, we will be focusing on emerging technologies because we are seeing businesses building totally new applications, and we want to help them. Third, we are focusing on how we can extend our world-class platform in terms of infrastructure and services so that more countries are able to adopt our Gen 2 Cloud infrastructure. We'll soon be making our Gen 2 Cloud available in India for those who develop and deploy applications while adhering to local regulations around data residency and security.

Q What, according to you, are the skills that Indian developers lack.

A: India is the global talent capital for IT, and Indian developers are a sought-after lot across the world. The enthusiasm Indian developers have to learn something new is great. Give them a problem and they will solve it. So, any skill shortage is addressed by their zeal to constantly innovate and improve. I don't see Indian developers lacking in any skill, per se.

Q What is your advice to developers who are working on the cloud and other cutting-edge technologies?

A: Someone once told me: "Not everything that is heralded as the next big thing is necessarily suitable for mass use." It is important for developers to always keep in mind some quantifiable metrics and real, business use cases. They should think about the user experience, mainly because in the customers' journey, the way their end users interact with an application becomes very crucial. Keeping up with the latest technologies, and trying to look at the bigger picture of what you are building and how it is useful to customers, end users and businesses — I think that does the trick! **END** 